

Cutsize and balance

Cutsize adds the weights of edges that cross the cut

The idea

- For every edge, if the endpoints sit in different parts, add its weight to the cut
- Balance ratio is the smaller part size over the larger
- Never celebrate a zero cut that parks every node on one side

Pseudocode

- Before any refinement, write the metrics
- Pseudocode sums cut edges for cutsize and part sizes for balance
- Imbalance is the absolute size gap over total size
- Open this module's examples file and find the Pseudocode section
- That written sketch is what you implement on the implement track and what the browser

Algorithm sketch

- Bad seed AE versus BCD cuts both heavy edges for cutsize twelve
- Golden ABC versus DE drops the cut to three with sizes three and two

Algorithm sketch — try these

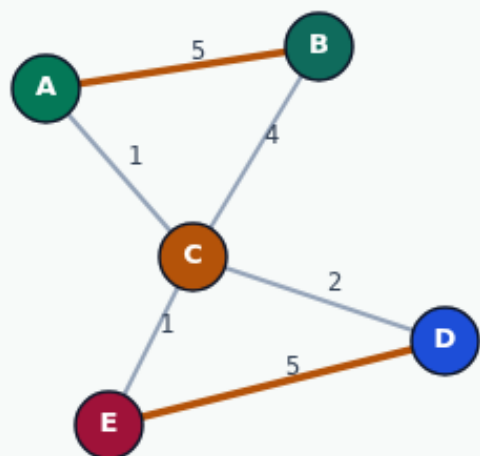
```
INPUT: assignment side[v], weighted edges
OUTPUT: cutsize, sizes, imbalance
cut  $\leftarrow \sum w(u,v)$  where side[u]  $\neq$  side[v]
size[p]  $\leftarrow \sum$  node_size on side p
imbalance  $\leftarrow |s_0 - s_1| / (s_0 + s_1)$ 
GOLDEN bad AE|BCD: cut=12, sizes 2|3
GOLDEN ABC|DE: cut=3, sizes 3|2
```

Same graph, two bipartitions

CUTSIZE & BALANCE

Same graph, two bipartitions

Step 1 / 5 · graph · module01-01-cutsizes-balance



Explanation

TINY_GRAPH has five cells. Partition quality is measured by cutsize (sum of cut edge weights) and balance (how even the two sides are).

- Cutsize = wire crossing the partition
- Balance ratio = min side / max side
- Both metrics matter in floorplanning

5 nodes, 6 edges
Target: legal 2-way split

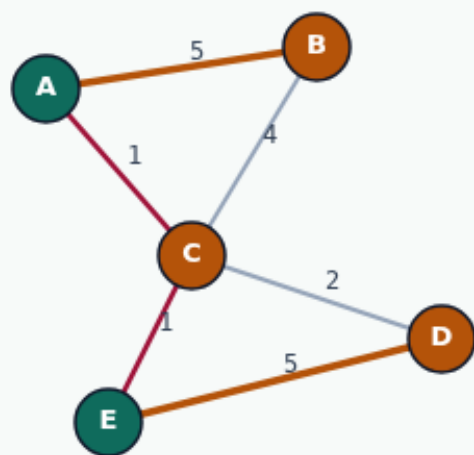
Same graph, two bipartitions

Bad seed: cutsize 12

CUTSIZE & BALANCE

Bad seed: cutsize 12

Step 2 / 5 · bad-seed · module01-01-cutsite-balance



Explanation

Seed AE|BCD cuts both heavy edges A-B(5) and D-E(5). Cutsite is 12 — a terrible wire cost even though the split is legal.

- Parts: A,E vs B,C,D
- Heavy internal edges become cut edges
- Same graph used across partitioning labs

```
seed:  
{ "A": "0", "E": "0", "B": "1", "C": "1", "D": "1" }  
cutsite: 12  
parts: AE|BCD
```

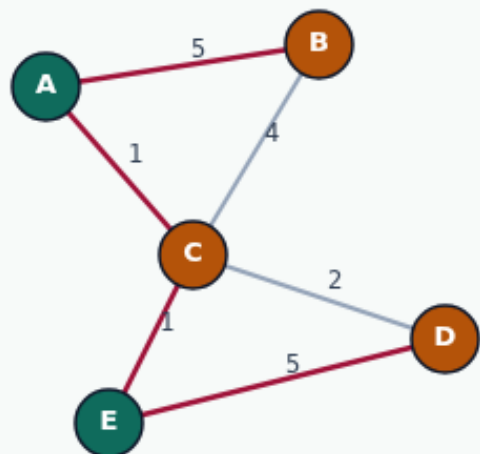
Bad seed: cutsite 12

Bad seed balance: ratio 2/3

CUTSIZE & BALANCE

Bad seed balance: ratio 2/3

Step 3 / 5 · bad-balance · module01-01-cutsizes-balance



Explanation

Side sizes are 2 vs 3, so balance ratio = $2/3$ and imbalance $|2-3|/5 = 0.2$. The split is moderately balanced — cutsize, not balance, is the problem.

- ratio = min/max of side sizes
- imbalance = $|s_0 - s_1| / n$
- Balance alone does not rank partitions

sizes: 2 vs 3
ratio: 0.6667
imbalance: 0.2

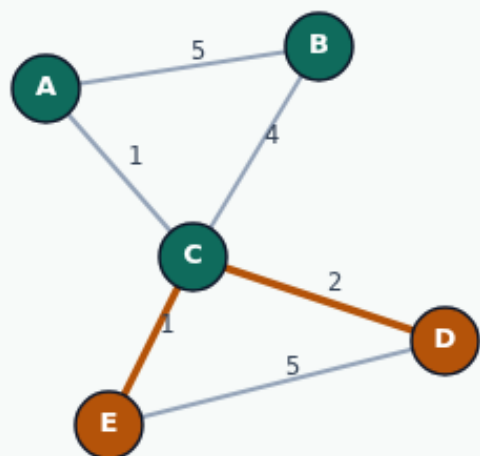
Bad seed balance: ratio 2/3

Golden ABC|DE: cutsize 3

CUTSIZE & BALANCE

Golden ABC|DE: cutsize 3

Step 4 / 5 · golden · module01-01-cutsite-balance



Explanation

Golden bipartition ABC|DE keeps A-B and D-E internal. Only weak bridges C-D(2) and C-E(1) are cut — cutsize drops to 3 with the same 2/3 ratio.

- Heavy edges stay inside communities
- Same balance ratio as bad seed
- Cutsite distinguishes quality

parts: ABC|DE
cutsite: 3
ratio: 2/3 (unchanged)

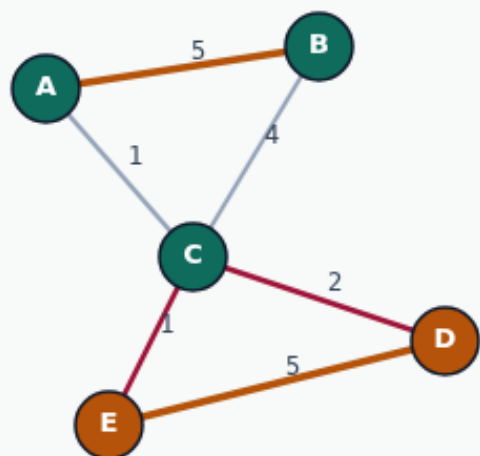
Golden ABC|DE: cutsize 3

Cutsizes + balance literacy

CUTSIZE & BALANCE

Cutsizes + balance literacy

Step 5 / 5 · takeaway · module01-01-cutsizes-balance



Explanation

Two partitions can share the same balance but differ wildly in cut. EDA flows optimize cut under balance constraints — never balance alone.

- Bad: AE|BCD cut 12, ratio 2/3
- Golden: ABC|DE cut 3, ratio 2/3
- Refinement labs repair bad seeds later

12 → 3 cut improvement
ratio stays 2/3

Browser lab track

- In the browser lab track, open the **cutsize-balance** lab from the tools shelf
- Load the starter graph, run the algorithm once
- Work the challenges that lock the goldens

Implement track

- In the implement track
- Parse the tiny graph, run the algorithm with a deterministic seed
- Match the browser goldens before you claim the checklist

Pitfalls

- Common traps
- For multilevel flows, verify coarsening before you blame the refiner

Your turn

- Complete the checklist for at least one track, preferably both
- Implement until your metrics match the starter goldens
- When you're ready, take the short quiz, then continue to the next module